
TEMPO: Regime-Aware Operator Learning with Locally Adapted POD Bases

Anonymous Author(s)

Affiliation

Address

email

Abstract

Operator learning methods typically approximate the solution map of a physical system with a single global model—an assumption that fails when solutions change character fundamentally across the parameter space.

We propose TEMPO: a framework that identifies latent dynamical regimes in snapshot data via Expectation-Maximization, builds a dedicated Neural-POD basis for each, and learns to route new inputs to the right regime at inference time. On multi-regime benchmarks, TEMPO achieves substantially lower errors than POD-DeepONet and vanilla DeepONet, with interpretable structure that reflects the underlying physics.

Keywords: Operator learning, DeepONet, Proper Orthogonal Decomposition, Expectation-Maximization, Gaussian mixture models, Regime discovery

1 Introduction

Operator learning addresses the approximation of mappings between infinite-dimensional function spaces, arising naturally in the solution of parametric partial differential equations (PDEs) governing physical systems [1]. This paradigm has demonstrated substantial promise across both theoretical analysis [1, 2] and practical applications including weather modeling, quantum systems, fluid and sound dynamics [3–6].

Among the most prominent architectures, Deep Operator Networks (DeepONet) [2] approximate operators via paired branch and trunk networks, where the branch encodes the input function and the trunk encodes the query location. The Fourier Neural Operator (FNO) [7] independently parameterizes the solution operator in the spectral domain, offering resolution-invariance and strong empirical performance. POD-DeepONet [8] improves upon standard DeepONet by replacing the learned trunk network with a fixed basis obtained from Proper Orthogonal Decomposition (POD) of training snapshots, yielding more compact and interpretable representations.

However, a single global POD basis is inherently limited for problems whose solutions exhibit qualitatively distinct dynamical regimes—such as solutions transitioning between laminar and turbulent behavior, or dispersive waves with parameter-dependent structure. In such settings, the optimal basis varies across the solution manifold, and a single linear subspace cannot represent the full solution diversity.

Several approaches enrich the POD basis to overcome this limitation. Sharma and Shankar [9] update the DeepONet trunk with multiple complementary bases—precomputed POD modes and a partition-of-unity mixture-of-experts (PoU-MoE) trunk, trained jointly with a single branch. On 2D and 3D problems with sharp spatial gradients, the POD-PoU ensemble achieves up to $2\times$ lower errors over POD-DeepONet [8] and $2\text{--}4\times$ over vanilla DeepONet.

Moreover, the linear structure of the POD basis is insufficient for strongly nonlinear solution manifolds. Mou et al. [10] propose Neural-POD, which constructs nonlinear orthogonal basis functions via iterative residual minimization analogous to Gram–Schmidt orthogonalization, representing each spatial mode and temporal component as neural networks trained sequentially on the residual variance.

Yet both approaches enrich the basis without identifying the underlying dynamical regimes present in the data. This idea has been pursued in the reduced order modeling (ROM) literature, where snapshot ensembles are partitioned into clusters and a separate local POD basis is constructed for each [11, 12]. More recently, Daniel et al. [13] extended this paradigm by clustering solution subspaces on the Grassmann manifold and training a deep neural network to select the appropriate local model from a dictionary at inference time. However, these methods operate offline on fixed snapshot sets, employ hard cluster assignments, and do not integrate regime discovery with operator learning in an end-to-end trainable architecture.

In this work, we propose **TEMPO** (*Regime-Aware Operator Learning with Locally Adapted POD Bases*). Our method integrates regime discovery directly into operator learning: a Gaussian mixture model over Neural-POD coefficient space identifies latent dynamical regimes via soft EM, constructing a dedicated locally adapted basis for each. At inference time, a learned gating network selects among regimes, producing predictions that are accurate across the full parameter space.

2 Problem Statement

The behavior of many physical systems is governed by a set of parameters $\kappa \in \mathcal{K} \subset \mathbb{R}^{d_\kappa}$: viscosity controls whether flow is laminar or turbulent, boundary forcing shapes wave patterns, material coefficients determine whether stress propagates smoothly or develops shocks. As κ varies across the parameter space $\mathcal{K}$, the corresponding solutions can change not just quantitatively but qualitatively, demanding fundamentally different representations in each regime. Predicting these solutions accurately and instantly, without re-running the underlying simulator, is the central problem we address.

Let $\Omega \subset \mathbb{R}^d$ be a bounded spatial domain and $\mathcal{T} \subset \mathbb{R}_{\geq 0}$ a time interval.

We are given a dataset of N solution snapshots:

$$\mathcal{D} = \left\{ (u^{(i)}, \kappa^{(i)}, t^{(i)}, s^{(i)}) \right\}_{i=1}^N, \quad s^{(i)}(\cdot) = s(\cdot, t^{(i)}; u^{(i)}, \kappa^{(i)}) \in L^2(\Omega), \quad (1)$$

where $u^{(i)} \in \mathcal{U} = L^2(\Omega)$ is the input function, $\kappa^{(i)} \in \mathcal{K}$ the physical parameter, $t^{(i)} \in \mathcal{T}$ the observation time, and $s^{(i)}$ the corresponding solution discretized on N_y quadrature points $\{(x_j, w_j)\}_{j=1}^{N_y}$ with norm $\|f\|^2 = \sum_j w_j f(x_j)^2$.

The dataset $\mathcal{D}$ captures $\mathcal{G}$ through a finite collection of snapshots; the underlying operator is defined as follows.

Definition 1 (Solution operator). *The solution operator*

$$\mathcal{G} : \mathcal{U} \times \mathcal{K} \rightarrow L^2(\Omega \times \mathcal{T})$$

maps each pair (u, κ) to the full space-time solution: $\mathcal{G}(u, \kappa) = s(\cdot, \cdot; u, \kappa)$.

The operator $\mathcal{G}$ is unknown and expensive to evaluate directly. The *operator learning problem* is to construct $\hat{\mathcal{G}} \approx \mathcal{G}$ from $\mathcal{D}$ alone, generalizing to unseen $(u, \kappa) \notin \mathcal{D}$ without solving the PDE.

The multi-regime assumption

When solutions change character fundamentally across $\mathcal{K}$, the optimal low-dimensional representation differs from regime to regime, and no single global basis captures the full solution diversity. We formalize this observation via a latent variable model over the snapshot ensemble.

Definition 2 (Generative model). *Each snapshot arises from one of M latent regimes. The regime label $z_i \in \{1, \dots, M\}$ is unobserved and drawn from a categorical prior,*

$$z_i \sim \text{Categorical}(\pi_1, \dots, \pi_M), \quad \pi_m \geq 0, \quad \sum_m \pi_m = 1. \quad (2)$$

Conditioned on $z_i = m$, the snapshot is distributed as

$$(s^{(i)} \mid z_i=m) \sim \mathcal{N}(\hat{s}_m^{(i)}, \sigma^2 W^{-1}), \quad (3)$$

where $W = \text{diag}(w_1, \dots, w_{N_y})$ is the quadrature weight matrix, $\sigma^2 > 0$ controls the softness of regime assignments, and $\hat{s}_m^{(i)}$ is the regime- m approximation to $s^{(i)}$.

The generative model leaves $\hat{s}_m^{(i)}$ unspecified. We require only that it admits a *separable low-rank form*:

$$\hat{s}_m^{(i)}(x) = \phi_0^{(m)}(x; \kappa^{(i)}) + \sum_{p=1}^{P_m} \phi_p^{(m)}(x; \kappa^{(i)}) \psi_p^{(m)}(t^{(i)}; \kappa^{(i)}), \quad (4)$$

where $\phi_p^{(m)}(\cdot; \kappa)$ are spatial basis functions and $\psi_p^{(m)}(\cdot; \kappa)$ temporal coefficients, both conditioned on κ . This form encompasses POD [8], kernel PCA [14], and learned neural bases [10]; in this work we adopt the latter.

Definition 3 (Regime approximator). *The parameters of regime m are $\Phi_m = \{\phi_p^{(m)}, \psi_p^{(m)}\}_{p=0}^{P_m}$, where $\phi_0^{(m)}(\cdot; \kappa)$ is a parameter-dependent mean, $\{\phi_p^{(m)}\}_{p \geq 1}$ spatial basis networks, $\{\psi_p^{(m)}\}_{p \geq 1}$ temporal coefficient networks, and $P_m \leq P_{\max}$ the adaptive mode count.*

The learning problem decomposes into two phases: discovering the latent regime structure from $\mathcal{D}$ *offline*, then training a deployable operator for fast inference *online*.

Phase 1: Offline regime discovery

Since the regime labels z_i in (3) are unobserved, we recover the bases $\{\Phi_m\}$, mixture weights π , and noise level σ^2 by maximizing the observed-data log-likelihood:

$$(\Phi^*, \pi^*, \sigma^{2*}) = \arg \max_{\Phi, \pi, \sigma^2} \sum_{i=1}^N \log \sum_{m=1}^M \frac{\pi_m \det(W)^{1/2}}{(2\pi\sigma^2)^{N_y/2}} \exp\left(-\frac{\|s^{(i)} - \hat{s}_m^{(i)}\|^2}{2\sigma^2}\right). \quad (5)$$

The solution yields soft regime assignments $\gamma_{im}^* = p(z_i=m \mid s^{(i)}, \Phi^*, \pi^*, \sigma^{2*})$ that encode the discovered regime structure of $\mathcal{D}$ and serve as supervision for Phase 2.

Phase 2: Online operator learning

With bases $\{\Phi_m^*\}$ and responsibilities $\{\gamma_{im}^*\}$ frozen, we train a *GatingNet* and M *BranchNets* that map a new input (u, κ) to a weighted combination of local regime predictions. The training objective balances three goals: reconstruction accuracy on $\mathcal{D}$, consistency of the learned gating with the offline regime structure $\{\gamma_{im}^*\}$, and diversity across regimes to prevent collapse. The precise architecture and loss are given in Section 7.

The quality of the final approximation is measured by the mean relative L^2 error on held-out test snapshots:

$$\varepsilon = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{\|s^{(i)} - \hat{s}(\cdot; u^{(i)}, \kappa^{(i)})\|}{\|s^{(i)}\|}. \quad (6)$$

The algorithm solves Phase 1 via EM and Phase 2 via gradient descent; details are in Section 7.

3 Theory

4 Computational Experiment

The goal of the experiment is to verify that the TEMPO framework can discover regime-specific NeuralPOD bases from multi-parameter PDE data and achieve accurate solution reconstruction on held-out samples.

Dataset. We use the PDEBench benchmark [15], specifically the 1D Burgers' equation $\partial_t u + u \partial_x u = \nu \partial_{xx} u$ on $x \in (-1, 1)$, $t \in (0, 2)$. The physical parameter $\kappa = \nu$ is varied across four values $\{0.001, 0.01, 0.1, 1.0\}$, producing solutions that range from smooth diffusive profiles ($\nu = 1.0$) to near-discontinuous shock structures ($\nu = 0.001$). Each viscosity file contains $N = 9,500$ samples on a uniform grid of $N_x = 1024$ spatial points and $N_t = 101$ time steps; the combined training set comprises approximately 3.8×10^6 snapshots.

PDEBench also provides several other benchmark tasks that we plan to evaluate in future work:

- **1D Advection** $\partial_t u + \beta \partial_x u = 0$: pure transport at speed β , tests resolution of travelling waves.
- **1D/2D Reaction–Diffusion**: two coupled fields (activator and inhibitor), produces Turing-type spatial patterns.
- **2D Darcy Flow** $-\nabla \cdot (a(x) \nabla u) = f$: steady-state elliptic equation, maps permeability field $a(x)$ to pressure u , widely used in subsurface flow modelling.
- **2D Shallow Water**: height and velocity fields over complex bathymetry, models flood dynamics and tsunami propagation.
- **1D/2D/3D Compressible Navier–Stokes**: three coupled fields (density, velocity, pressure), most challenging task in the benchmark, tests high-Reynolds turbulent regimes.

Metrics. Model accuracy is evaluated using four standard scalar metrics computed on a held-out test split (10% of samples):

- **MSE**: mean squared error $\frac{1}{N} \sum_i \|\hat{u}_i - u_i\|_2^2$ and **RMSE**;
- **Relative L^2 error**: $\|\hat{u} - u\|_2 / \|u\|_2$;
- **MAPE**: mean absolute percentage error $\frac{1}{N} \sum_i |\hat{u}_i - u_i| / (|u_i| + \epsilon)$.

Expected results. Table 1 will report these four metrics for TEMPO and the Neural-POD baseline [10] across the two most contrasting viscosity regimes. Figure 2 will show training loss curves, residual norm decay per mode, and space–time reconstruction heatmaps for both regimes.

Table 1: Reconstruction accuracy on 1D Burgers' test set.

Method	MSE	RMSE	Rel. L^2	MAPE
DeepONet	–	–	–	–
POD-DeepONet	–	–	–	–
Neural-POD	–	–	–	–
TEMPO	–	–	–	–

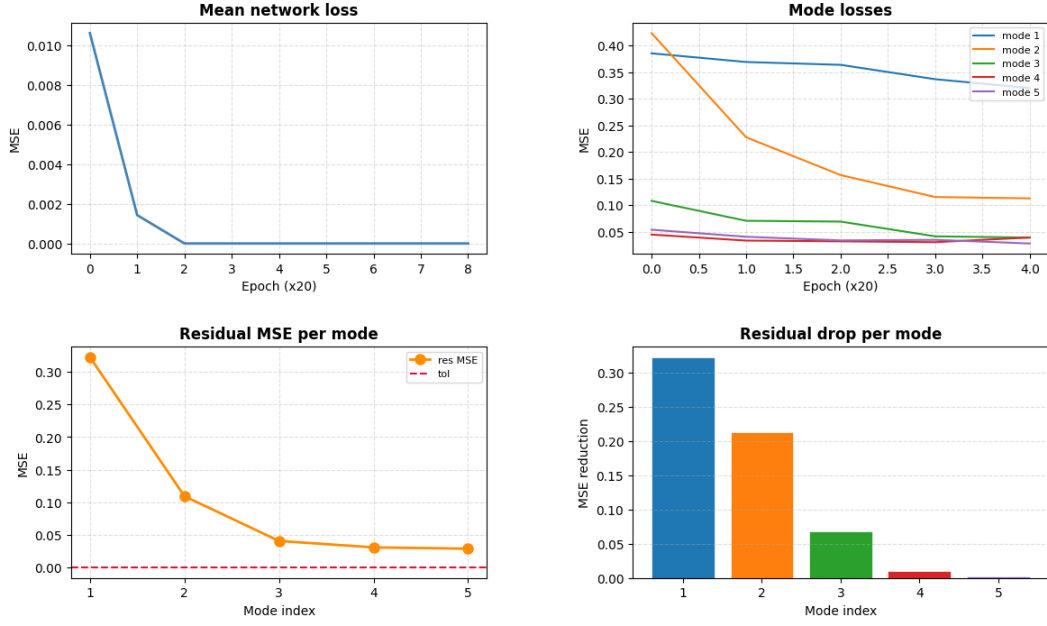


Figure 1: Experimental results $\nu = 1.0$.

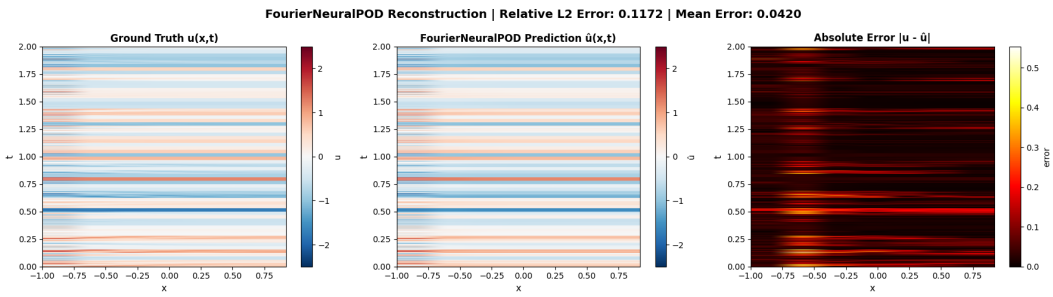


Figure 2: Experimental results $\nu = 1.0$.

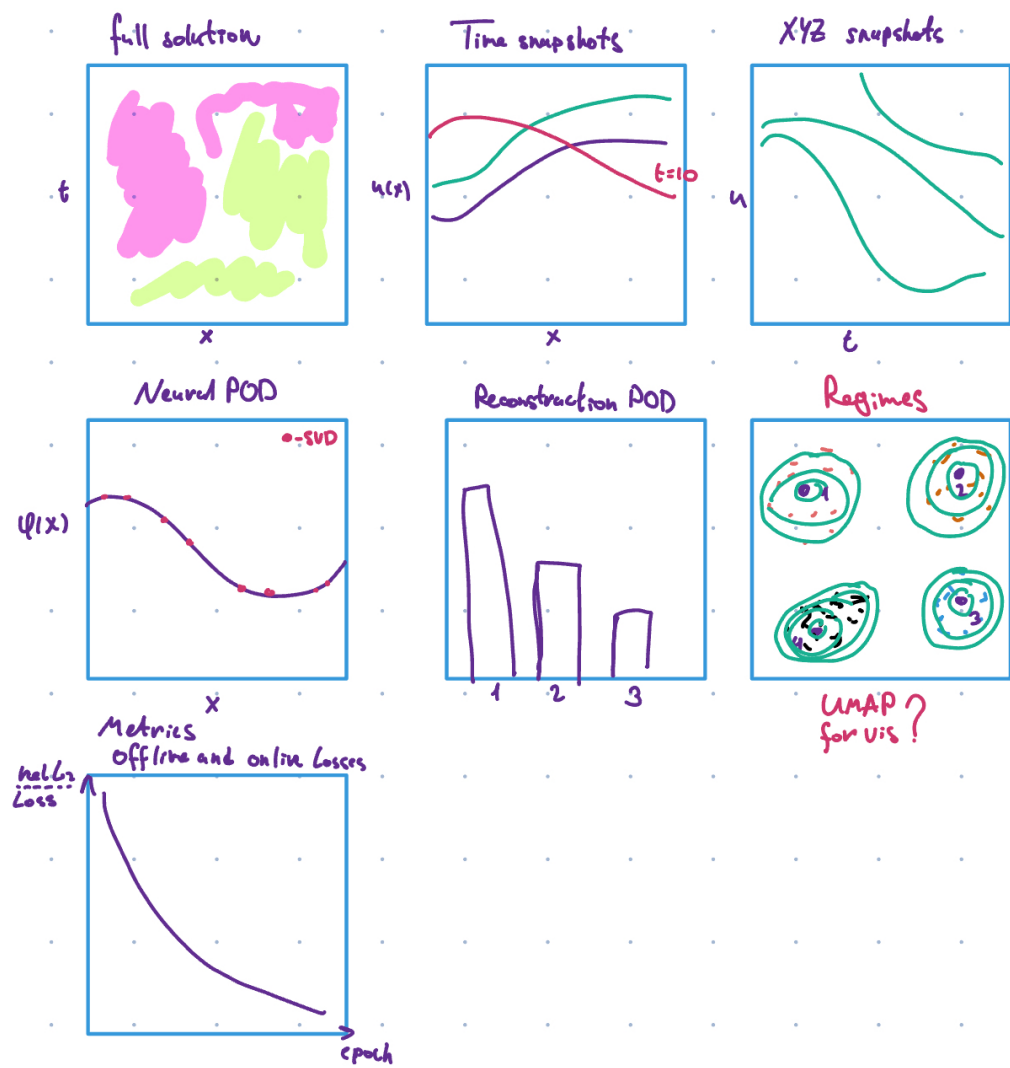


Figure 3: Ideas for plots.

TODOOO

METHOD OR STATEMENT????????????????

Definition 4 (TEMPO predictor). *Given input function u sampled at sensor locations and parameter κ , the TEMPO prediction at query point $x \in \Omega$ is*

$$\hat{s}(x; u, \kappa) = \sum_{m=1}^M w_m(u, \kappa) \left[\Phi_0^{(m)}(x; \kappa) + \sum_{k=1}^{P_m} b_{m,k}(u, \kappa) \Phi_k^{(m)}(x; \kappa) \right], \quad (7)$$

where $w_m(u, \kappa) \in [0, 1]$, $\sum_m w_m = 1$, are the GatingNet outputs, and $b_{m,k}(u, \kappa)$ are the Branch-Net expansion coefficients for regime m

Online

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_{i=1}^N \left\| s^{(i)} - \hat{s}(\cdot; u^{(i)}, \kappa^{(i)}) \right\|^2, \quad (8)$$

$$\mathcal{L}_{\text{KL}} = \frac{1}{N} \sum_{i,m} w_m^{(i)} \log \frac{w_m^{(i)}}{\gamma_{im}^*}, \quad (9)$$

$$\mathcal{L}_{\text{ent}} = -\frac{1}{N} \sum_{i,m} w_m^{(i)} \log w_m^{(i)}, \quad (10)$$

combined as

$$(W^*, B^*) = \arg \min_{W, B} \mathcal{L}_{\text{online}}, \quad \mathcal{L}_{\text{online}} = \mathcal{L}_{\text{data}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} - \lambda_H \mathcal{L}_{\text{ent}}, \quad (11)$$

Outline. The rest of the paper is organized as follows...

5 Related Work

Topic #1. TODO

Topic #2. TODO

6 Preliminaries

6.1 General notation

In this section, we introduce the general notation used in the rest of the paper and the basic assumptions.

6.2 Assumptions

TODO

7 Method

8 Experiments

To verify the theoretical estimates obtained, we conducted a detailed empirical study...

9 Discussion

TODO

10 Conclusion

TODO

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A Appendix / supplemental material

A.1 Additional experiments / Proofs of Theorems

TODO